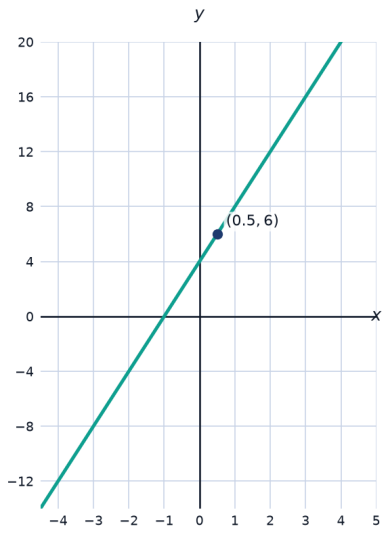
Coordinate Geometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

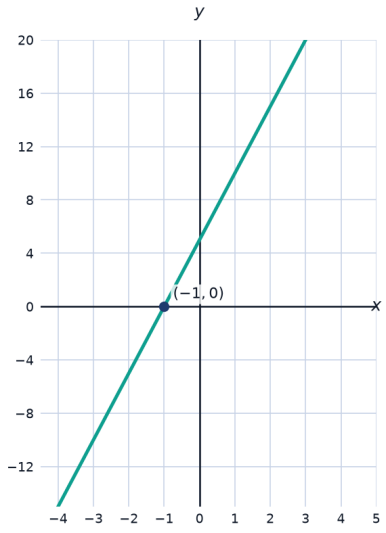
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 4x + 4$ in converting between fractions and decimals, evaluate $f(5)$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 5$ in recurring decimals to fractions, evaluate $f(2)$.



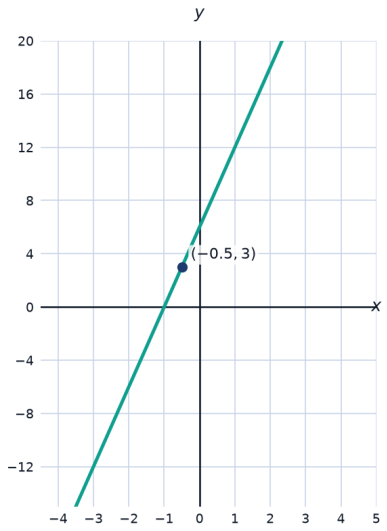
3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for recurring decimals to fractions consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 6x + 6$ in multiplying with decimals, evaluate $f(2)$.

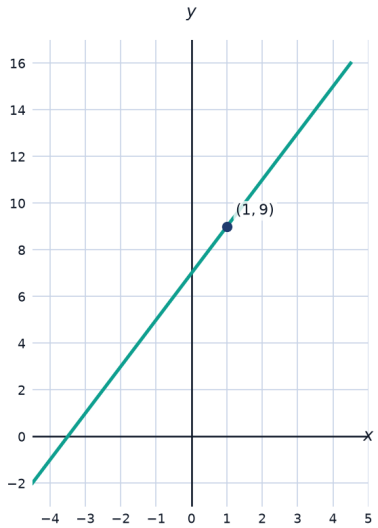


5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for multiplying with decimals consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 2x + 7$ in dividing with decimals, evaluate $f(2)$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for dividing with decimals consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Coordinate Geometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Show that the points $A(1, 3)$, $B(4, 7)$ and $C(7, 3)$ form an isosceles triangle.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Geometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Two similar cylinders have radii 4 cm and 6 cm. If the volume of the smaller is 160π cm³, find the volume of the larger.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

If $f(x) = 4x - 3$ and $g(x) = x^2 + 1$, find (a) $gf(2)$ and (b) an expression for $g^{-1}(x)$ for $x \geq 1$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Using the same bag (5 red, 3 blue), find the probability of drawing exactly one red ball.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Functions

[Mensuration · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Number

[Number · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

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17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 10.01 Sets & Venn Diagrams

[Sets & Venn Diagrams · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Algebra & Graphs

[Probability & Statistics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Mensuration

[Transformations & Vectors · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

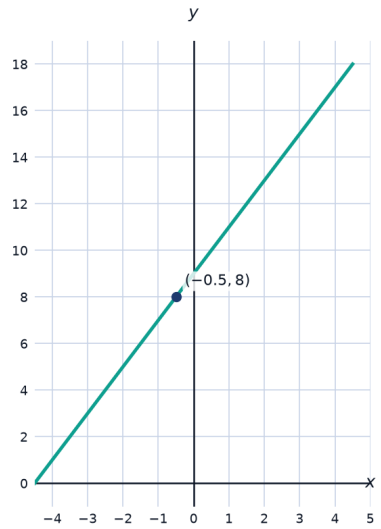
20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Geometry

[Trigonometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 9$ in decimals and place value, evaluate $f(2)$.



a. 13

b. 15

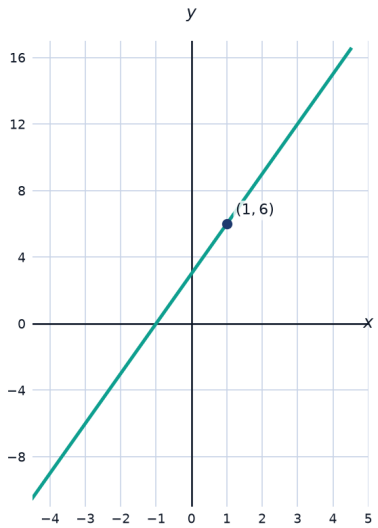
c. 4

d. 17

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 3$ in adding and subtracting decimals, evaluate $f(2)$.



a. 9

b. 12

c. 6

d. 13

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for decimals and place value consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for adding and subtracting decimals consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

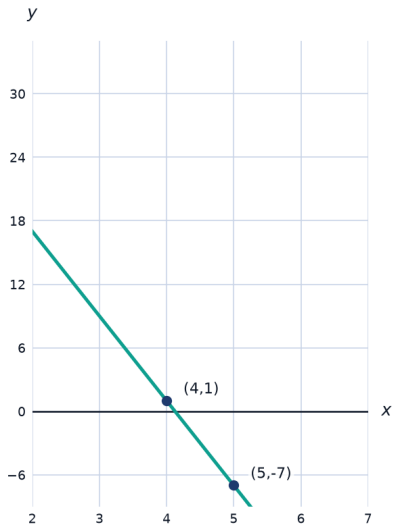
c. 19.5

d. 21.5

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the equation of the line through $(4, 1)$ and $(5, -7)$.



a. $y = -8x + 31$

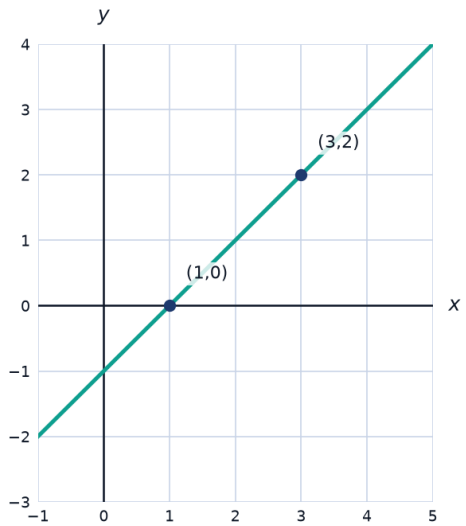
b. $y = -8x + 35$

c. $y = -8x + 33$

d. $y = -8x - 33$

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find the gradient of the line joining $(1, 0)$ and $(3, 2)$.



a. 2

b. 1

c. 0

d. -1

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27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A line has gradient 3. What is the gradient of a line perpendicular to it?

a. $2/3$

b. $-1/3$

c. $-13/3$

d. 3

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the distance between $(0, 0)$ and $(6, 8)$.

a. 3

b. 10

c. 6

d. 13

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A line has gradient -5. What is the gradient of a line parallel to it?

a. -1

b. -5

c. -4

d. $1/5$

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the equation of the line with gradient 3 passing through $(-2, -3)$.

a. $y = 3x + 3$

b. $y = 3x + 5$

c. $y = 3x - 3$

d. $y = 3x + 1$

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Area of trapezium with parallel sides 5 and 9, height 4:



a. 28

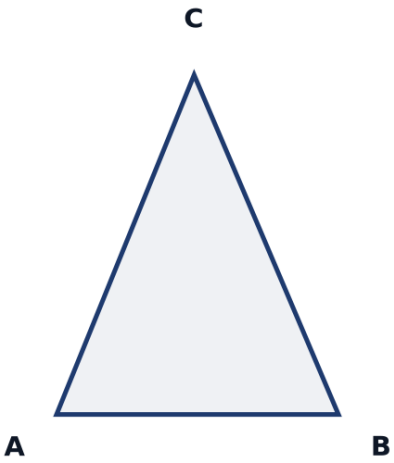
b. 56

c. 36

d. 14

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In a triangle, two angles are 68° and 67°. Find the third angle.



a. 45°

b. 52°

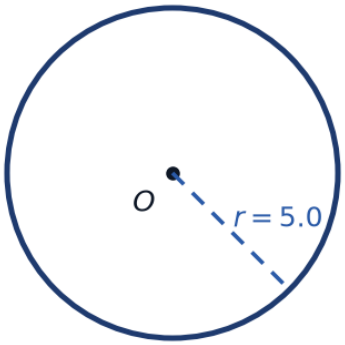
c. 57°

d. 56°

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In a circle, a tangent meets the radius at:



a. 30°

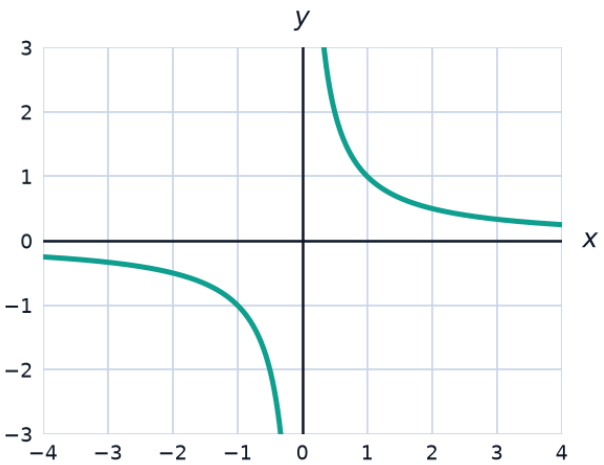
b. 45°

c. 60°

d. 90°

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Angle in a semicircle is always:



a. 45°

b. 90°

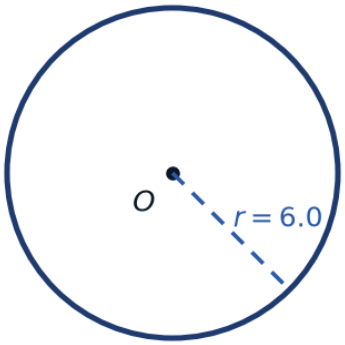
c. 180°

d. 60°

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A sector has radius 6 and angle 60°. Arc length is:



a. 2π

b. 4π

c. π

d. 6π

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Points P, Q, R lie on a circle with centre O. Tangents from external point T touch the circle at P and Q. If angle $\angle PTQ = 46^\circ$, what is the size of angle $\angle PRQ$ on the major arc?

a. 67°

b. 74°

c. 134°

d. 92°

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Points A, B, and C lie on a circle with centre O. Angle $\angle AOB = 110^\circ$. What is the size of angle $\angle ACB$ on the major arc?

a. 55°

b. 70°

c. 110°

d. 125°

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Algebra & Graphs

$ABCD$ is a cyclic quadrilateral. Angle $\angle DAB = 78^\circ$ and angle $\angle ABC = 104^\circ$. What is the size of angle $\angle BCD$?

a. 76°

b. 102°

c. 104°

d. 78°

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Mensuration

Two tangents TP and TQ are drawn from an external point T to a circle with centre O. If angle PTQ = 46°, find angle POQ.

a. 134°

b. 46°

c. 144°

d. 90°

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Geometry

Two chords AB and CD of a circle intersect at an interior point P. If $AP = 4$ cm, $PB = 9$ cm, and $CP = 3$ cm, what is the length of PD?

a. 10 cm

b. 12 cm

c. 14 cm

d. 16 cm

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Trigonometry

A line TA is tangent to a circle at A. ABC is a triangle inscribed in the circle. If angle $\angle TAB = 52^\circ$, what is the size of angle $\angle ACB$?

a. 38°

b. 52°

c. 104°

d. 128°

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 7.01 Transformations & Vectors

Two tangents TP and TQ are drawn from an external point T to a circle with centre O. If angle PTQ = 50°, find angle TPQ.

a. 65°

b. 50°

c. 75°

d. 55°

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 8.01 Probability & Statistics

What is the size of each exterior angle of a regular octagon (8-sided polygon)?

a. 45°

b. 60°

c. 135°

d. 30°

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Functions

A rhombus has diagonals of lengths 10 cm and 24 cm. What is the perimeter of the rhombus?

a. 52 cm

b. 48 cm

c. 68 cm

d. 34 cm

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Number

What is the sum of the interior angles of a convex hexagon (6-sided polygon)?

a. 360°

b. 540°

c. 720°

d. 900°

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